MedTwin

From Lab Reports to an Explainable Patient Digital Twin

Personalized health insights powered by a patient digital twin, machine learning, guideline-grounded RAG, and Azure AI.

EDUCATIONAL DEMO · NOT A MEDICAL DIAGNOSTIC SYSTEM

The Problem

Why generic AI chat isn't enough for diabetes care

Confusing lab reports

Patients get HbA1c, blood pressure, and cholesterol values with no plain-language explanation of what they mean.

Generic chatbots fall short

Tools like ChatGPT answer basic questions, but aren't grounded in clinical guidelines or the patient's own data.

No personalized risk insight

There's no easy way to see how your numbers compare to real population data or medical targets.



Reports full of numbers.
Little clarity.

MedTwin turns raw values into a clear, personal, and cited answer.

Concepts Used & Our Solution

Six building blocks working together, not a single chatbot guessing at your labs

Digital Twin

Your real values, structured into one profile

Risk Scoring

Estimates glycemic risk from NHANES data

Guideline Answers

Grounded in ADA, NICE & WHO guidance

Grounded Reasoning

Reasons only over your data and evidence

Verified Pipeline

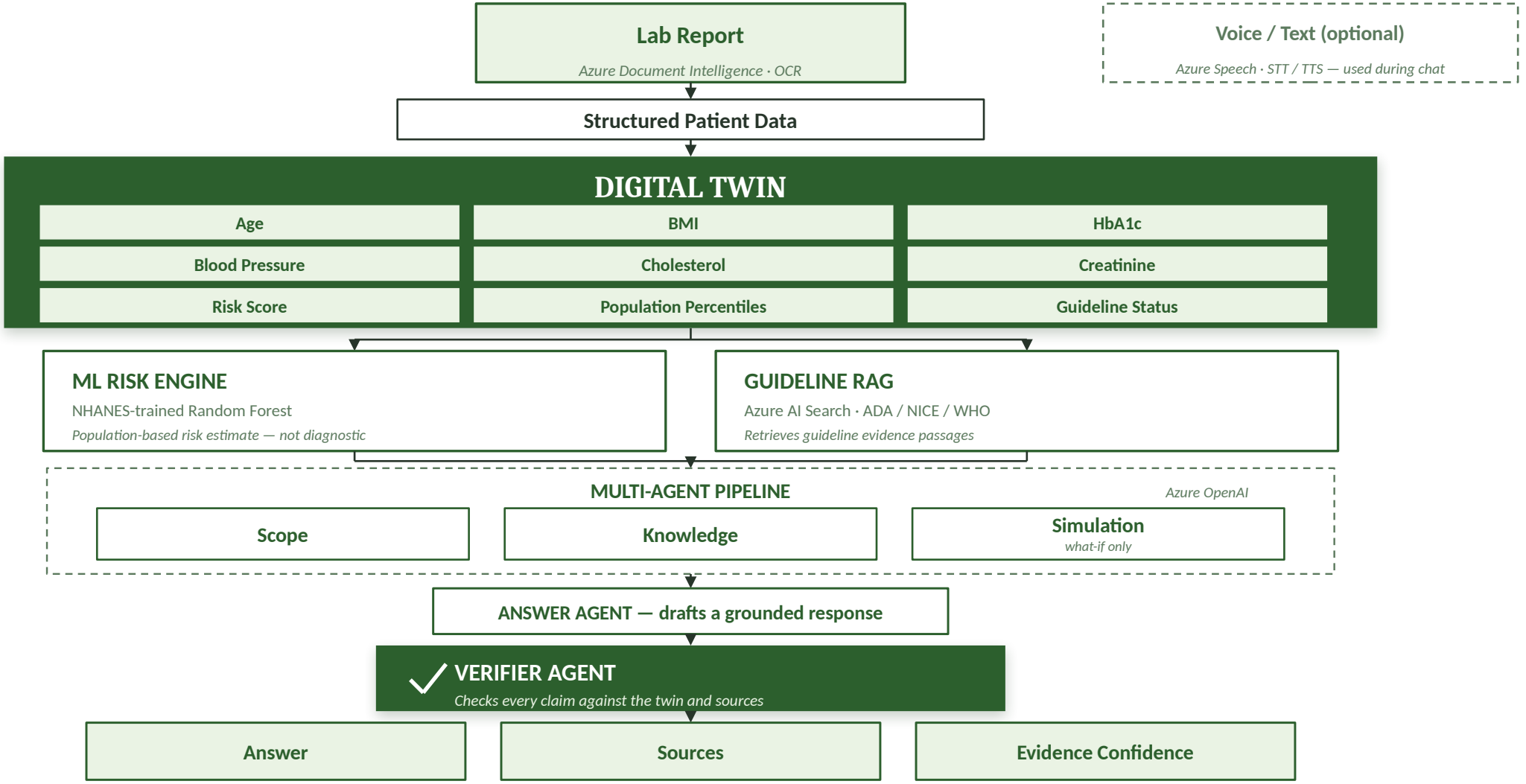
Scope → Knowledge → Simulate → Verify

Built on Azure

OCR, Search, Speech & Cosmos DB

MedTwin — End-to-End Architecture

From patient data to a verified, evidence-grounded response



Limitations & Future Scope

Limitations

- Trained on a survey snapshot, not longitudinal patient records
- What-if results are population estimates, not medical forecasts
- Scoped to adults diagnosed at age 30+, a proxy for Type 2
- Confidence score reflects evidence support, not clinical correctness

Future Scope

- Track patient progress over time with a longitudinal twin
- Expand guideline coverage: NICE, WHO, ADA/EASD consensus
- Add a clinician-mode dashboard and multi-language support
- Full production deployment on Azure App Service

Thank You

MedTwin — Deliverables

From Lab Reports to an Explainable Patient Digital Twin

Team Members

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1. Working Prototype

https://gentle-tree-0d75a2100.1.azurestaticapps.net

2. GitHub Repository

https://github.com/RobinChahal0010/MedTwin

3. Project Presentation Video

https://youtu.be/f--Fs5kWGm8